

Top Trading Cycles in a practical setting

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Abstract

This thesis explores the Top Trading Cycles algorithm in the practical setting of the allocations of GPs, General Practitioner, in Norway for people wanting to switch or get a GP. It explores how different focus of *good* results can change both the runtime and results, and how to make the algorithm run as efficiently as possible. This thesis proposes an implementation of the Top Trading Cycles that can reduce the waitlists for GPs in Norway by over 80%.

Acknowledgements

I would like to thank my supervisor Fredrik Manne for his guidance and support throughout this project. I would also like to thank Juni Weisteen Bjerde for input during this project.

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1. Introduction

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1.1. Motivation

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1.2. Contributions

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1.3. Thesis Outline

The remainder of this thesis is organized as follows:

- **Section 2** provides background on the Top Trading Cycles mechanism and its application to GP allocation.
- **Section 3** formally defines the computational problems arising from TTC as variants of cycle cover problems on directed graphs.
- **Section 4** describes our Rust implementation and the algorithmic strategies employed.
- **Section 5** presents experimental results comparing different priority strategies.
- **Section 6** concludes and discusses future work.

2. Background

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2.1. The GP Allocation Problem in Norway

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2.2. Top Trading Cycles

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2.3. Related Work

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3. Problem Formalization

In this chapter we try to define the GP allocation problem and variations of it. In addition we go through how to construct a graph given a GP allocation problem and finally how we can reduce the GP allocation problem to the Cycle Cover problem in directed graphs.

3.1. GP allocation problem

3.1.1. Input

Consider the GP allocation problem as given a list of patients P and a list of doctors D . We then have patients $p_1, p_2, \dots, p_n$ and doctors $d_1, d_2, \dots, d_m$. We also are given the current and preferred doctor assignments as arrays:

- $D_{\text{current}}[i]$ denotes the index of the current doctor assigned to patient p_i (or $\perp$ if unassigned)
- $D_{\text{preferred}}[i]$ denotes the index of the preferred doctor for patient p_i

In addition we are given a priority function $R : P \rightarrow \mathbb{N}$, mapping some positive integer to each patient, the higher the number the higher the priority.

3.1.2. Feasible solution

A feasible solution for the GP allocation problem is a subset of patients $S \subseteq P$ such that the directed graph

$$G_S = (S, E_S), E_S = \{(a, b) \mid a, b \in S, D_{\text{preferred}}[\text{idx}(a)] = D_{\text{current}}[\text{idx}(b)]\} \quad (1)$$

consists of a vertex-disjoint union of directed cycles that cover all vertices in S . Where $\text{idx}(x)$ is a function giving the index of a patient x . So this means our solution S is a set of all patients that can exchange doctor in one or more cycles.

3.1.3. Optimization criterion

With this feasible solution defined we can start defining variants where we want feasible solution that maximizes some *score*. When choosing what patients should be exchanging we might have different qualities that we want in our solutions. We might want a solution that exchanges as many patients as possible or that exchanges patients with the highest priority.

First we define our general ordering

Definition 3.1.3.1 (Optimization criterion): An ordering $\succ$ over feasible solutions. A solution is optimal if it is maximal under $\succ$.

Then we can build upon this to create our two variants

3.1.3.1. Lexicographic maximization by priority

Definition 3.1.3.1.1 (Sorted priority vector): Given a feasible solution $S \subseteq P$ the sorted priority vector is:

$$\pi(S) = \text{sort}((R(p)_{p \in S})) \quad (2)$$

where the vector is sorted in decreasing order.

This vector we can then use to compare solutions and order them lexicographically by priority.

Definition 3.1.3.1.2 (Lexicographic ordering by priority):

$$S \succ_{\text{lex}} S' \text{ iff } \pi(S) \text{ is lexicographically larger than } \pi(S') \quad (3)$$

In addition if $\pi(S')$ is a strict prefix of $\pi(S)$ then $\pi(S)$ is greater.

We can then using this, if given two solution S and S' determine which one is *better* by the lexicographic ordering by priority.

Example (Ordering two solutions lexicographically by priority):

We use the example graph G below as the graph to create solutions from.

Given solutions

1. $S = \{4, 5\}$ represents the subgraph G_S containing cycle $P_4 \rightarrow P_5 \rightarrow P_4$
2. $S' = \{1, 2, 3\}$ represents the subgraph $G_{S'}$ containing cycles $P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_1$

For simplicity's sake we say that the $R(x) = \text{idx}(x)$, so in S the priority values are the same as S .

Then $\pi(S) = [5, 4]$ and $\pi(S') = [3, 2, 1]$ and we can see that S is then greater by the ordering $\succ_{\text{lex}}$. Because at the first point they differ, $S[0] \neq S'[0]$, $S[0]$ is greater.

So for our first variant we want to find the maximal solution under the $\succ_{\text{lex}}$ ordering. This means always prioritizing those with highest priority.

3.1.3.2. Maximizing cardinality

The other variant is finding a solution with the largest cardinality, e.g. the solution that contains the most patients.

Definition 3.1.3.2.1 (Ordering by cardinality):

$$S \succ_{\text{size}} S' \text{ iff } |S| > |S'| \quad (4)$$

This solution will be the one that exchanges the most patients and therefore makes the most amount of people happy.

Example (Ordering two solutions by cardinality):

Say we have graph G as in the figure below. Then our solution sets $S, S' \subseteq V(G)$

Given solutions

1. $S = \{4, 5\}$ represents the subgraph G_S containing cycle $P_4 \rightarrow P_5 \rightarrow P_4$
2. $S' = \{1, 2, 3\}$ represents the subgraph $G_{S'}$ containing cycles $P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow P_1$

By then the ordering $\succ_{\text{size}} |S| = 4, |S'| = 3$, so S is greater than S' under $\succ_{\text{size}}$. The absolute maximal solution under $\succ_{\text{size}}$ would be $\{1, 2, 3, 4, 5\}$ since that solution would contain both cycles.

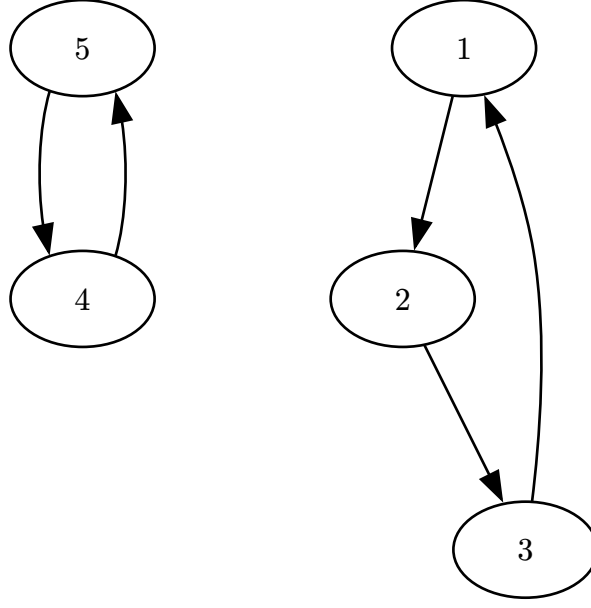


Figure 1: An example graph G .

4. Implementation

4.1. TTC

First I started implementing the Top Trading cycles implementation described in Huitfeldt's paper. For this algorithm we are given the lists of patients and doctors P, D and $D_{\text{cur}}, D_{\text{pref}}$ and a priority function $R : P \rightarrow \mathbb{N}$, mapping some positive integer to each patient, the higher the number the higher the priority. We create our graph

Let $I = \{0, \dots, |P| - 1\}$. Then:

$$G = (V, E), \quad V = P \cup D, \quad E = \{(p_i, D_{\text{pref}[i]}) \mid i \in I\} \cup \{(D_{\text{cur}[i]}, p_i) \mid i \in I\}$$

So we use the patients and doctors as nodes and create an edge from a patient to its preferred doctor and a doctor to all patient with that doctor as its current doctor.

Then the algorithm goes as follows:

```
1 let resolved_patients = []
2 let p_prio = Sorted list of patients by priority
3 let wants_to_switch = [true] * |P|
4 for each p ∈ p_prio do
5   if let cycle = dfs_find_cycle(G, R, p) do
6     for each p ∈ cycle do
7       let i = idx(p)
8       wants_to_switch[i] = false
9       resolved_patients.push(p)
10    end
11  end
12 end
13 return resolved_patients
```

The function “dfs_find_cycle” starts a dfs from patient p and tries to find a cycle containing p . When the dfs enters a doctor node d and it can choose between multiple patients to go to, it always starts by exploring the patient p with highest priority $R(p)$.

While we can argue that this approach at least cares about the priority of a patient, we cannot guarantee that it always chooses the “best” solution. In that the greedy dfs might ruin for cycles that we could get without breaking any of the other cycles.

Consider this example graph:

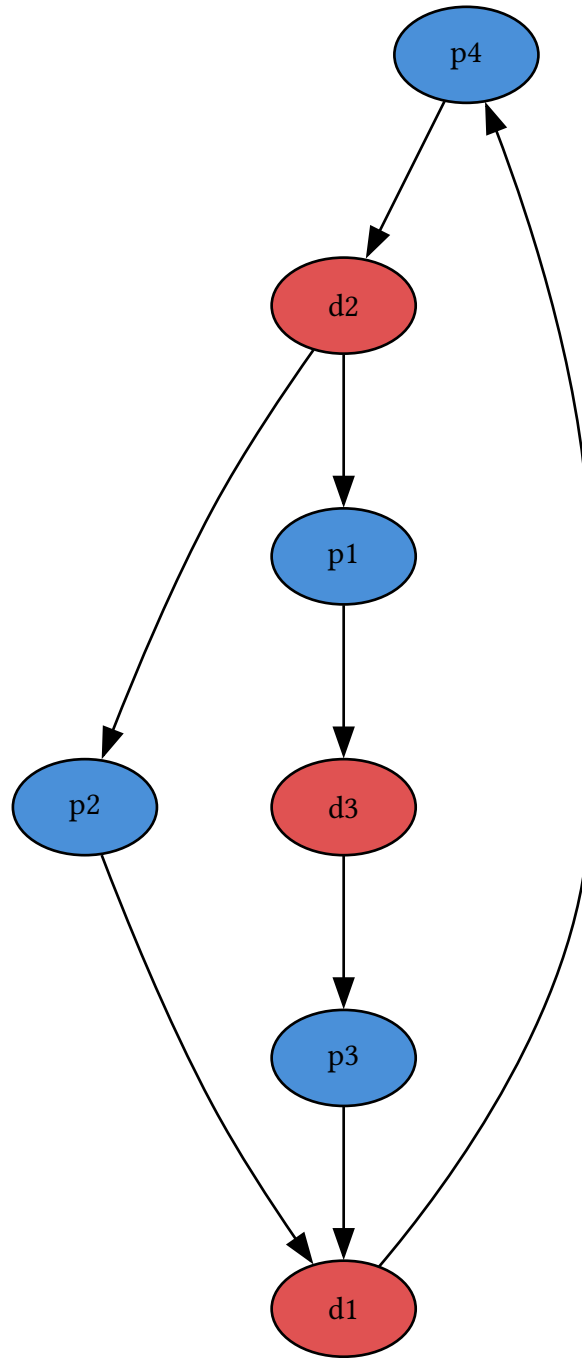


Figure 2: Example graph G where algorithm would choose suboptimal solution. Numbers are priority values.

4.2. Exact algorithm for maximizing total switches

4.3. Metaheuristics

5. Experimental Results

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5.1. Experimental Setup

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5.2. Performance Comparison

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5.3. Impact of Priority Strategies

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6. Conclusion

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6.1. Summary

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6.2. Future Work

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